A Verification Note on the Path-Compression Formulation of the Micali–Vazirani Pointer-Machine Question

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Abstract

This note repairs, rather than endorses, a proposed negative solution to the open problem about maintaining the iterated bud map bud* in one phase of the Micali–Vazirani matching algorithm on a pointer machine. The outcome is *partial progress, not a solution.* I do not obtain a proof or disproof of the open problem in Vazirani's 2024 paper. The negative theorem in the earlier draft is not justified by the public literature: it depends on an unpublished lower-bound claim cited in a 2013 arXiv version and later removed from the 2020 arXiv and 2024 journal versions.

The unconditional contribution here is a precise reduction of the raw path-compression implementation to standard union–find with naive linking. If a phase creates b bud elements and performs s relevant parent-pointer operations on them, then raw path compression costs

$$O\Big(s \log_{s/b+2} b\Big).$$

Consequently, using the standard MV accounting $s = O(m)$, raw path compression is already linear in every phase with $m \geq b^{1+\varepsilon}$ for any fixed $\varepsilon > 0$, and in particular in dense phases with $m \geq n^{1+\varepsilon}$. Thus any remaining obstruction to a linear raw path-compression bound must lie in the sparse or near-sparse regime and must use more MV-specific structure than laminarity of petals alone. A separate section records the exact conditional statement that would follow if the unpublished Pettie–Vazirani lower bound were supplied.

1 Status of the repaired note

The relevant question in Vazirani's journal paper is whether the operation bud*, needed during DDFS in a phase of the Micali–Vazirani (MV) algorithm, admits a linear-time pointer-machine implementation. The paper records two related approaches:

(a) the raw path-compression approach attributed to Micali and Vazirani;

(b) the possibility that Tarjan's set-union implementation, on the special operation sequences generated by an MV phase, may have linear total cost.

These are related but they are not the same data-structural statement. A lower bound for raw path compression would not by itself refute a more powerful union–find implementation, and a linear bound for Tarjan union–find would not prove that raw path compression has the same bound.

The earlier draft asserted a negative solution by quoting a 2013 arXiv sentence saying that Pettie and Vazirani had an infinite family of phases in which raw path compression takes $\Omega(m\alpha(m,n))$ time. The reference listed there was only "in preparation". Since the lower-bound construction and proof are not present in the cited source, the assertion cannot be used as an unconditional theorem. Moreover, later versions of the same line of work return to saying that the question is open. Therefore the corrected status is:

> No solution is obtained here. The note gives a rigorous parameterized upper bound for raw path compression, a dense-regime linear corollary, a clarification of what a genuine negative solution would need to prove, and a warning that laminarity alone cannot be the missing reason for linearity.

2 The raw bud-parent data structure

Fix one phase of MV on a graph $G = (V, E)$, and write $|V| = n$, $|E| = m$. During DDFS, the algorithm needs the fixed point of the current bud map,

$$\mathrm{bud}^*(v) = \mathrm{bud}(\mathrm{bud}(\cdots \mathrm{bud}(v) \cdots)).$$

The part relevant to the present note is the parent-pointer forest on bud elements. When a previously constructed petal is encountered from an ordinary vertex, the implementation first follows the stored pointer from that vertex to the corresponding petal node and then to the bud of the petal; this ordinary vertex-to-bud work is charged to graph edges in the usual MV accounting. The remaining operation is a representative query on the current bud-parent forest.

Definition 1 (Raw bud path compression)**.** A *raw bud path-compression implementation* is an implementation with the following operations on the bud elements created or used in a phase.

- A new bud element starts as a singleton root.
- When the MV phase makes one current bud representative subordinate to another current bud representative, the data structure performs a naive link: the parent of one current root is set to the other prescribed root. The direction of the link is dictated by the MV bud relation, not by rank, size, index, or randomization.
- A bud^*-query follows parent pointers to the current root and then rewrites every pointer on the traversed path to point directly to that root.

The cost is the number of pointer traversals and pointer updates in these operations, up to a constant factor.

This definition intentionally separates raw path compression from Tarjan's set-union algorithm. Tarjan's algorithm is free to maintain rank or size information and to choose a linking direction that need not preserve the visible nesting tree of petals. The raw implementation is not.

3 A verified black-box upper bound

The following theorem is simply the Tarjan–van Leeuwen analysis of path compression with naive linking, translated into the notation of one MV phase.

Theorem 2 (Parameterized raw path-compression bound)**.** *Consider one MV phase. Suppose the raw bud path-compression structure uses $b \geq 2$ bud elements and performs $s \geq b$ total relevant operations on those elements, including singleton creations, links, and* bud^**-queries. Then the total pointer-machine cost of these operations is*

$$T_{\mathrm{pc}}(s, b) = O\Big(s \log_{s/b+2} b\Big) .$$

In particular, $T_{\mathrm{pc}}(s, b) = O(s \log b)$.

Proof. After the ordinary vertex-to-bud steps charged to graph edges are removed, the remaining structure is exactly a union–find forest with naive linking and full path compression. Singleton bud creation is a make-set operation. Each time one current representative becomes subordinate to another prescribed current representative, the operation is a naive link. Each bud*-query is a find with full path compression.

Tarjan and van Leeuwen proved that any intermixed sequence of s such operations on b elements has total cost $O(s \log_{s/b+2} b)$. The theorem follows by applying this result to the operation sequence generated by the phase. The displayed $O(s \log b)$ bound is the special case obtained from the fact that the logarithm base is at least 3 when $s \geq b$. □

Corollary 3 (Standard fallback bound for a phase)**.** *Using the standard MV per-phase accounting that the number of relevant* bud*-*operations is* $O(m)$*, raw path compression costs* $O(m \log n)$ *per phase. More precisely, if* b *is the number of bud elements used in the phase and* $s = O(m)$*, then the cost is*

$$O\Big(s \log_{s/b+2} b\Big).$$

Proof. Theorem 2 gives the parameterized statement. Since the bud elements are graph vertices used as buds, $b \leq n$, and since $s = O(m)$ in the usual phase accounting, the crude bound $O(s \log b)$ gives $O(m \log n)$. □

4 A dense-regime linear corollary

The parameterized form of Theorem 2 gives a small but useful unconditional linearity result. It says that the possible obstruction to linear raw path compression is confined to phases in which the number of bud elements is close to the number of relevant operations.

Theorem 4 (Linearity when the graph is dense relative to its bud set)**.** *Fix* $\varepsilon > 0$*. There is a constant* C_ε *such that the following holds for every MV phase. If the phase uses* b *bud elements, performs* $s \leq C_0 m$ *relevant raw path-compression operations, and*

$$m \geq b^{1+\varepsilon},$$

then the raw path-compression cost in the phase is at most $C_\varepsilon C_0 m$*. In asymptotic notation, the cost is* $O_\varepsilon(m)$*.*

Proof. By Theorem 2, for a universal constant C,

$$T_{\mathrm{pc}}(s, b) \leq C s \log_{s/b+2} b.$$

Let $x = s/b \geq 1$. We split into two cases.

If $x \geq b^{\varepsilon/2}$, then $\log(x+2) \geq (\varepsilon/2) \log b$, and hence

$$T_{\mathrm{pc}}(s, b) \leq \frac{2C}{\varepsilon} s \leq \frac{2CC_0}{\varepsilon} m.$$

If $1 \leq x < b^{\varepsilon/2}$, then the crude logarithm-base lower bound $\log(x+2) \geq \log 3$ gives

$$T_{\mathrm{pc}}(s, b) \leq \frac{C}{\log 3}\, bx \log b \leq \frac{C}{\log 3}\, b^{1+\varepsilon/2} \log b.$$

For all sufficiently large b, $\log b \leq b^{\varepsilon/2}$, so this is at most a constant depending only on ε times $b^{1+\varepsilon} \leq m$. The finitely many smaller values of b are absorbed by enlarging the constant. This proves the claim. □

Corollary 5 (Dense graph phases)**.** *For every fixed $\varepsilon > 0$, raw path compression is linear per phase on all phases with $m \geq n^{1+\varepsilon}$, assuming the standard $s = O(m)$ phase accounting.*

Proof. The bud elements are vertices, so $b \leq n$. Therefore $m \geq n^{1+\varepsilon}$ implies $m \geq b^{1+\varepsilon}$, and Theorem 4 applies. $\square$

Remark 6. This dense-regime corollary does not solve the open problem. It only shows that a counterexample to linear raw path compression, if one exists, must occur in the sparse or near-sparse regime, where the number of edge scans is not polynomially larger than the number of active bud elements.

5 What a genuine negative solution would need

The uploaded draft tried to prove a negative answer from an asserted Pettie–Vazirani lower bound. The following proposition is the logically correct conditional version of that argument.

Proposition 7 (Conditional negative implication)**.** *Suppose there is an infinite family of MV phases Φ_i, with graph edge counts m_i, such that the raw bud path-compression implementation performs at least*

$$c\, m_i f_i$$

pointer operations in phase Φ_i, for some universal constant $c > 0$ and for some numbers $f_i \to \infty$. Then raw path compression is not an $O(m)$-time implementation of bud* *in all MV phases.*

Proof. If raw path compression were always linear, then there would be a constant C such that every phase with m edges had cost at most Cm. Choose i with $cf_i > C$. The lower bound on Φ_i then gives cost greater than Cm_i, a contradiction. $\square$

Remark 8 (Why the 2013 sentence is not enough). The 2013 arXiv version states, without giving the construction, that Pettie and Vazirani had an infinite family on which path compression in a phase takes $\Omega(m\alpha(m, n))$ time, and its bibliography lists the source only as "in preparation". To turn that sentence into a proof of Proposition 7, a paper would need to supply at least the following data:

(i) the actual graph and starting matching family;

(ii) the exact MV tie-breaking or an argument that every allowed tie-breaking gives the same bad operation sequence;

(iii) the induced sequence of bud links and bud*-queries;

(iv) a lower-bound proof for raw path compression on that exact sequence;

(v) verification that the lower-bound factor tends to infinity along the family, not merely that an inverse-Ackermann expression appears formally.

Without these ingredients, the negative statement is an unverified citation, not a mathematical proof.

6 Why laminarity alone cannot be the missing argument

One might hope that the laminar nesting of MV petals alone forces the union–find sequence to be easier than a worst-case sequence. The following basic observation shows that laminarity by itself is too weak: every ordinary union–find history is laminar.

Proposition 9 (Laminarity is automatic for set-union histories). *Consider any sequence of disjoint-set union operations starting from singleton sets, with arbitrary finds interspersed. Let $\mathcal{H}$ be the collection of all sets that ever exist immediately after a make-set operation or a successful union operation. Then $\mathcal{H}$ is laminar: for any $A, B \in \mathcal{H}$, either $A \cap B = \varnothing$, or $A \subseteq B$, or $B \subseteq A$.*

Proof. A set, once created, changes only by being merged with a disjoint current set to form a larger set. It is never split. We prove the claim by induction over the operation sequence. Initially the singleton sets are pairwise disjoint. Find operations do not change the represented sets. A successful union replaces two disjoint current sets X and Y by $Z = X \cup Y$. Every old historical set is either disjoint from both X and Y, contained in X, contained in Y, or contains one of the current sets involved in the union. In each case its relation to the new set Z is disjointness, containment in Z, or containment of Z. Thus laminarity is preserved. □

Corollary 10. *A proof that an MV phase has linear raw path-compression or Tarjan set-union cost cannot rely only on the fact that petals/blossoms form a laminar family. It must use additional MV-specific restrictions, such as tenacity ordering, DDFS supports, predecessor-edge structure, synchronization of search levels, or some stronger property of the generated operation sequence.*

Proof. Worst-case union–find executions also generate laminar histories by Proposition 9. Therefore laminarity alone does not distinguish MV-generated sequences from arbitrary set-union sequences. □

7 Conclusions

The corrected conclusions are as follows.

Claim	Status in this note
Raw path compression is always linear in an MV phase	Not proved here; remains open from the present note.
Raw path compression has a public, verified $\Omega(m\alpha(m,n))$ MV lower bound	Not established; the public 2013 citation lacks the construction and proof.
Raw path compression has a verified fallback bound	Proved via Tarjan–van Leeuwen: $O(s \log_{s/b+2} b)$, hence $O(m \log n)$ under standard MV accounting.
Raw path compression is linear in dense phases	Proved: $O_\varepsilon(m)$ if $m \geq b^{1+\varepsilon}$, in particular if $m \geq n^{1+\varepsilon}$.
Tarjan set union is linear on all MV phases	Not proved here; this is a separate residual problem.
Laminarity explains possible linearity	False as a standalone explanation; laminarity is automatic for all union histories.

Thus the repaired note is a partial-progress and verification note. It does not solve Vazirani's problem. It does, however, remove the unsupported negative claim, isolate the exact hypothesis needed for such a claim, and prove that the open difficulty can be confined to sparse or near-sparse phases and to MV-specific structure beyond mere laminar nesting.

References

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